Semiring-Based Linear Algebra Framework for Two-Way Regular Path Queries

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Abstract. Two-way regular path queries (2-RPQs) allow the use of regular languages over edges and their inverses in edge-labeled graphs to constrain paths of interest. Adopted in modern graph query languages such as GQL, SQL/PGQ, and Cypher, 2-RPQs have become part of the GQL ISO standard. However, their efficient evaluation on large-scale graphs remains challenging. In this paper, we propose a semiring-based generalization of the linear algebra approach for evaluating single-source 2-RPQs. The original LARPQ algorithm solves the reachability problem using Boolean matrices. We generalize it by replacing Boolean algebra with arbitrary semirings, enabling a unified solution for various path-aggregation problems. Specifically, we employ the path semiring to find all simple paths and all trails. The algorithms are implemented using the SuiteSparse:GraphBLAS library within the LAGraph infrastructure. Experimental evaluation on real-world knowledge graphs demonstrates significant performance improvements over state-of-the-art solutions.

Keywords: Regular path queries · Two-way RPQ · Linear algebra · Semirings · Graph databases · Sparse matrices.

1 Introduction

Language-constrained path querying (FLPQ) [4] provides a mechanism for searching paths in edge-labeled graphs where constraints are specified in terms of formal languages. A path is considered valid if the word formed by concatenating the labels of its edges belongs to the specified language. When constraints are regular languages, such queries are called regular path queries (RPQs). In practice, it is often useful to allow traversal opposite to edge directions, leading to two-way regular path queries (2-RPQs) [8, 6]. Queries specifying only a start or an end vertex are known as single-source or single-destination queries, respectively. Both have been adopted in graph query languages such as GQL, SQL/PGQ [11], and SPARQL [1].

Graph database systems support different path query semantics, as the choice significantly affects both result interpretation and computational complexity.

Common semantics include: reachability (all vertices reachable via any valid path), all-shortest-paths (one shortest path per reachable vertex), all-simple-paths (all simple paths satisfying the query), and all-trails (all trails satisfying the query) [10]. Each semantics serves different use cases, and the query engine must select the appropriate algorithm accordingly.

Efficient evaluation of RPQs and 2-RPQs remains an active research area [9, 13]. Sparse linear algebra based methods have shown promise for RPQ evaluation [3]. In particular, the LARPQ (Linear Algebra-based Regular Path Query) algorithm [5] employs a breadth-first traversal that simultaneously processes both the graph and the finite automaton representing the regular language. While LARPQ demonstrates promising performance, it was designed solely for reachability, and its application to queries with fixed destination vertices remains unexplored.

The contributions of this paper are as follows.

- A formal generalization of the LARPQ algorithm to a semiring-based framework, enabling the solution of various path-aggregation problems within a unified algorithmic structure.
- Instantiation of the framework with specific semirings to solve distinct problems: (i) Boolean semiring for reachability, and (ii) path semiring with index unary operators for all-simple-paths and all-trails semantics.
- ??? Proof of correctness and computational complexity analysis for the generalized algorithm.
- Experimental evaluation on real-world knowledge graphs demonstrating significant performance improvements over state-of-the-art alternatives.

2 Preliminaries

In this section, we introduce the foundational concepts required throughout the paper, including edge-labeled graphs, paths, finite automata, and regular path queries.

2.1 Edge-Labeled Graphs

We begin by defining the graph data model used in this work.

Definition 1 (Edge-labeled graph). *A quadruple $G = \langle V, E, L, \lambda_G \rangle$ is called an edge-labeled graph (or simply a graph) if:*

- *V is a finite set of vertices;*
- *$E \subseteq V \times V$ is a finite set of edges;*
- *L is a finite set of labels;*
- *$\lambda_G : E \mapsto 2^L$ represents edge labels.*

For the remainder of this paper, we assume that vertices of any graph $G = \langle V, E, L, \lambda_G \rangle$ are enumerated, and without loss of generality, $V = \{1, 2, \ldots, |V|\}$.

To enable traversal in both directions along edges, we introduce the notion of inverted edges and labels. Let $G = \langle V, E, L, \lambda_G \rangle$ be an edge-labeled graph. We define:

- $a^- \notin L$ for each $a \in L$, with $(a^-)^- = a$ and $a = b \Leftrightarrow a^- = b^-$;
- $e^- = (v, u)$ where $e = (u, v)$;
- $E^- = \{e^- \mid e \in E\}$;
- $L^- = \{a^- \mid a \in L\}$;
- $\lambda_G^- : E^- \mapsto 2^{L^-}$, where $\lambda_G^-(e^-) = \{a^- \mid a \in \lambda_G(e)\}$.

We also define the bidirectional edge and label sets:

- $E^{\leftrightarrow} = E \cup E^-$;
- $L^{\leftrightarrow} = L \cup L^-$;
- $\lambda_G^{\leftrightarrow} : E^{\leftrightarrow} \mapsto 2^{L^{\leftrightarrow}}$, where $\lambda_G^{\leftrightarrow}(e) = \lambda_G(e) \cup \lambda_G^-(e)$;
- $G^{\leftrightarrow} = \langle V, E^{\leftrightarrow}, L^{\leftrightarrow}, \lambda_G^{\leftrightarrow} \rangle$.

The graph $G^{\leftrightarrow}$ allows traversal along both original and inverted edges, which is essential for evaluating two-way regular path queries.

2.2 Paths

Definition 2 (Path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ of length n in the graph $G = \langle V, E, L, \lambda_G \rangle$ from u_1 to v_n is a finite sequence of edges $e_i = (u_i, v_i) \in E$ such that $\forall 1 \leq i \leq n-1 : v_i = u_{i+1}$.*

We also consider zero-length paths, represented by an empty sequence from a vertex to itself.

Definition 3 (Simple path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is simple if it contains no repeated vertices except possibly the first and last. Formally, assuming $e_i = (v_i, u_i)$ that means $\forall 1 \leq i, j \leq n : v_i = u_j \Rightarrow j = i + 1$.*

Definition 4 (Trail). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is a trail if it contains no repeated edges. Formally, $\forall 1 \leq i, j \leq n : e_i = e_j \Rightarrow i = j$.*

Definition 5 (Two-way path). *A two-way path (2-path) $\pi = (e_1, e_2, \ldots, e_n)$ in G is a path in $G^{\leftrightarrow}$. Two-way simple paths and two-way trails are defined analogously.*

The mapping ω associates paths with words over the label alphabet:

- $\omega_G(\pi) = \{a_1 \cdot a_2 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G(e_i)\}$ for a path $\pi = (e_1, e_2, \ldots, e_n)$ in G;
- $\omega_{G^{\leftrightarrow}}(\pi) = \{a_1 \cdot a_2 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G^{\leftrightarrow}(e_i)\}$ for a 2-path $\pi = (e_1, e_2, \ldots, e_n)$ in G.

Here, $\cdot$ denotes string concatenation.

2.3 Two-Way Finite Automata

Regular languages, recognized by finite automata, provide the foundation for specifying path constraints. For two-way path queries, we require a variant of nondeterministic finite automata.

Definition 6 (Two-way NFA). *A two-way nondeterministic finite automaton (2-NFA) is a tuple $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F \rangle$, where:*

- *Q is a finite set of states;*
- *Σ is a finite alphabet;*
- *$\Delta \subseteq Q \times Q$ is the transition relation;*
- *$\lambda_N : \Delta \mapsto 2^{\Sigma^{\leftrightarrow}}$ maps transitions to labels;*
- *$Q_S \subseteq Q$ is the set of starting states;*
- *$Q_F \subseteq Q$ is the set of final states.*

Definition 7 (Two-way DFA). *A two-way deterministic finite automaton (2-DFA) is a special case of 2-NFA with $|Q_S| = 1$ and $\forall (q, q'), (q, q'') \in \Delta : q' \neq q'' \Rightarrow \lambda_N((q, q')) \cap \lambda_N((q, q'')) = \varnothing$.*

A 2-NFA can be viewed as a graph $G_N = \langle Q, \Delta, \Sigma^{\leftrightarrow}, \lambda_N \rangle$ equipped with sets of starting and final states. Conversely, a graph $G = \langle V, E, L, \lambda_G \rangle$ equipped with $V_S \subseteq V$ can be seen as a 2-NFA $N_G = \langle V, L, E, \lambda_G^{\leftrightarrow}, V_S, V \rangle$ that accepts:

$$[[N_G]] = \bigcup_{\text{2-path } \pi_G \text{ in } G \text{ from } v_S \in V_S} \omega_{G^{\leftrightarrow}}(\pi_G)$$

2.4 Two-Way Regular Path Queries

Definition 8 (Two-way regular path query). *A two-way regular path query (2-RPQ) is a triple $\langle G, R, v_s \rangle$ where $G = \langle V, E, L, \lambda_G \rangle$ is a graph, R is a regular language over the alphabet $L^{\leftrightarrow}$, and $v_s \in V$ is a starting vertex.*

We define several semantic variants of 2-RPQs that are considered in this paper:

- **Single-source reachability (SSR):**

$$[[Q]]_{SSR} = \{u \in V \mid \exists \text{ 2-path } \pi \text{ in } G \text{ from } v_s \text{ to } u \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **Single-destination reachability (SDR):**

$$[[Q]]_{SDR} = \{u \in V \mid \exists \text{ 2-path } \pi \text{ in } G \text{ from } u \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **Single-source all simple paths (SSASP):**

$$[[Q]]_{SSASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **Single-destination all simple paths (SDASP):**

$$[[Q]]_{SDASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

– **Single-source all trails (SSAT):**

$$[[Q]]_{SSAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

– **Single-destination all trails (SDAT):**

$$[[Q]]_{SDAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

3 Linear Algebra, Graphs And Relations

In this section, we establish the linear algebra foundations required for our approach. We introduce the concept of semirings, discuss matrix representations of relations, and present the original LARPQ algorithm for solving the reachability problem.

3.1 Semirings

Semirings provide an algebraic framework that generalizes both Boolean algebra and path-based computations. A semiring consists of a set equipped with two binary operations that satisfy certain axioms.

Definition 9 (Semiring). *A semiring is a tuple* $\langle D_{out}, D_{in_1}, D_{in_2}, \oplus, \otimes, \mathbf{0}\rangle$ *where:*

- D_{out}*,* D_{in_1}*, and* D_{in_2} *are three domains*
- $\langle D_{out}, \oplus, \mathbf{0}\rangle$ *is a commutative monoid with an identity element* $\mathbf{0} \in D_{out}$*;*
- $\langle D_{out}, D_{in_1}, D_{in_2}, \otimes\rangle$ *is a closed binary operator;*

In graph processing algorithms, semirings are defined differently from algebraic semirings due to the versatility of having different domains and irrelevancy of distributivity and multiplicative identity elements. Two specific semirings are particularly important for this work: the Boolean semiring for reachability and the path semiring for enumerating paths.

3.2 Matrix Representations

Relations between finite sets can be represented using matrices. For two enumerated finite sets $A = \{1, 2, \ldots, n\}$ and $B = \{1, 2, \ldots, m\}$, a binary matrix T of size $n \times m$ can represent a relation $T \subseteq A \times B$ where $T_{ij} = 1 \Leftrightarrow (i, j) \in T$.

Let $G = \langle V, E, L, \lambda_G\rangle$ be an arbitrary graph and $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F\rangle$ an arbitrary 2-NFA. For each label $a \in L^{\leftrightarrow}$, we define:

- $E_a = \{e \mid e \in E, a \in \lambda_G^{\leftrightarrow}(e)\}$;
- $\Delta_a = \{\delta \mid \delta \in \Delta, a \in \lambda_N(\delta)\}$.

Definition 10 (Adjacency matrix of a label). *Let* G_a *be the matrix representing the binary relation* E_a*. Then* G_a *is called the adjacency matrix of label* a*.*

Definition 11 (Boolean decomposition). *A Boolean decomposition of the adjacency matrix* G *is the set of Boolean matrices* $\{G_a \mid a \in L^{\leftrightarrow}\}$*.*

3.3 Boolean Matrix Operations

For matrices over the Boolean semiring $\langle \mathbb{B} = \{0, 1\}, \vee, \wedge, \neg, 0, 1\rangle$, we define the following operations. Let $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$ be matrices:

- **Sum:** $A_{n\times m} + B_{n\times m} = (a_{ij} \vee b_{ij})_{n\times m}$;
- **Product:** $A_{n\times m} \times B_{m\times k} = (\bigvee_{l=1}^{m} a_{il} \wedge b_{lj})_{n\times k}$;
- **Transposition:** $A^T_{n\times m} = (a_{ji})_{m\times n}$;
- **Zero-matrix:** $0_{n\times k} = (0)_{n\times k}$;
- **Mask:** $A_{n\times m}\langle C_{n\times m}\rangle = (a_{ij} \wedge c_{ij})_{n\times m}$;
- **Complement:** $\neg A = (\neg a_{ij})$.

These operations suffice for expressing the reachability algorithm.

3.4 The Path Semiring

For solving path enumeration problems (all simple paths, all trails), we require a more expressive semiring based on path representations.

Let $\text{RP}\textsc{aths} = 2^{\mathbb{N}^*}$, where $\pi' = (v_1, v_2, \ldots, v_n) \in \mathbb{N}^*$ represents a 2-path $\pi = ((v_1, v_2), (v_2, v_3), \ldots, (v_{n-1}, v_n))$ of length $n - 1$. For $a, b \in \text{RP}\textsc{aths}$ and $c \in \mathbb{B}$, we define:

- **Sum:** $a \oplus b = a \cup b$;
- **Product:** $a \otimes c = \begin{cases} a & \text{if } c = 1 \\ \varnothing & \text{otherwise} \end{cases}$.

To this end, we can introduce a semiring for deciding more complex path problems:

- $\textsc{Paths} = \langle \text{RP}\textsc{aths}, \text{RP}\textsc{aths}, \mathbb{B} = \{0, 1\}, \cup, \mathbf{fst}, \varnothing\rangle$ where $\mathbf{fst}(x, y) = \begin{cases} x & \text{if } y = 1 \\ \varnothing & \text{otherwise} \end{cases}$

Let $A = (a_{ij})$, $B = (b_{ij})$ be matrices over $\text{RP}\textsc{aths}$ and $C = (c_{ij})$ be a Boolean matrix. We extend the operations to matrices:

- **Sum:** $A_{n\times m} \oplus B_{n\times m} = (a_{ij} \oplus b_{ij})_{n\times m}$;
- **Product:** $A_{n\times m} \otimes C_{m\times k} = (\bigoplus_{l=1}^{m} a_{il} \otimes c_{lj})_{n\times k}$;
- **Zero-matrix:** $0^{\textsc{Paths}}_{n\times k} = (\varnothing)_{n\times k}$.

Definition 12 (boolean decomposition). *A Boolean decomposition of the adjacency matrix G is the set of Boolean matrices $\{G_a \mid a \in L^{\leftrightarrow}\}$.*

3.5 Index Unary Operators

The path semiring alone is insufficient for enforcing path semantics (simple paths, trails). We introduce index unary operators that can filter paths based on vertex or edge repetition.

Definition 13 (Index unary operator). *Let $A_{n\times m} = (a_{ij})$ be a matrix over a semiring R. An index unary operator $I : R\times\mathbb{N}\times\mathbb{N} \mapsto R$ is applied element-wise: $I(A) = (I(a_{ij}, i, j))_{n\times m}$.*

For matrices over RPATHS, we define specific index unary operators:

- $I_{Simple}(a, i, j) = \{(v_1, v_2, \ldots, v_n, j) \mid (v_1, v_2, \ldots, v_n) \in a, v_1 \neq v_n$ and $\forall 2 \leq t \leq n : v_t \neq j\}$;
- $I_{Trail}(a, i, j) = \{(v_1, v_2, \ldots, v_n, j) \mid (v_1, v_2, \ldots, v_n) \in a$ and $\forall 1 \leq t \leq n-1 : (v_t, v_{t+1}) \neq (v_n, j)\}$.

The I_{Simple} operator extends paths while filtering out those that would create vertex repetitions (enforcing simple paths). The I_{Trail} operator filters out extensions that would create edge repetitions (enforcing trails).

3.6 The Original LARPQ Algorithm for Reachability

We now present the original LARPQ algorithm for solving single-source 2-RPQ reachability, which serves as the foundation for our semiring generalization.

Input: $\mathcal{N} = \langle Q, \Sigma, \Delta, \lambda_{\mathcal{N}}, Q_S, Q_F\rangle$, $\mathcal{G} = \langle V, E, L, \lambda_{\mathcal{G}}\rangle$, $v_s \in V$
Output: Vector P^F of size $1 \times |V|$

1. $\{N^a\} \leftarrow$ Boolean decomposition of the $\mathcal{N}$ adjacency matrix
2. $\{G^a\} \leftarrow$ Boolean decomposition of the $\mathcal{G}$ adjacency matrix
3. $M_{|Q|\times|V|} \leftarrow \mathbf{0}_{|Q|\times|V|}$
4. $P_{|Q|\times|V|} \leftarrow \mathbf{0}_{|Q|\times|V|}$
5. For each $q \in Q_S$: set $M_{qv_s} \leftarrow 1$
6. While $M \neq \mathbf{0}$:
   - $M \leftarrow \bigoplus_{a\in\Sigma^{\leftrightarrow}\cap L^{\leftrightarrow}}((M^T \times N^a)^T \times G^a)\langle\neg P\rangle$
   - $P \leftarrow P \oplus M$
7. $F_{1\times|Q|} \leftarrow \mathbf{0}_{1\times|Q|}$
8. For each $q \in Q_F$: set $F_{1q} \leftarrow 1$
9. Return $P^F = F \times P$

Fig. 1: Single-Source 2-RPQ Reachability using Linear Algebra

The algorithm maintains two matrices: M (the frontier) represents the relation between automaton states and graph vertices reached in the current step, while P accumulates all reachable pairs. The key insight is that on each iteration, $M_{qv} = 1$ if and only if there exists a 2-path in $\mathcal{G}$ from v_s to v and a path in $\mathcal{N}$ from some $q_s \in Q_S$ to q, both having the same labels and of length n (the iteration number). The mask $\langle\neg P\rangle$ ensures that each vertex-automaton state pair is processed only once, preventing infinite loops.

The correctness of the algorithm follows from the simultaneous breadth-first traversal of both the graph and the automaton. After termination, $P_{q_F v} = 1$ for

$q_F \in Q_F$ if and only if there exists a 2-path from v_s to v whose label sequence belongs to the language accepted by $\mathcal{N}$.

The algorithm can also solve single-destination reachability by reversing the query. Let $\mathcal{R}^- = \{w^- = a_n^- a_{n-1}^- \dots a_1^- \mid w = a_1 a_2 \dots a_n \in \mathcal{R}\}$ and $\mathcal{N}^- = \langle Q, \Sigma, \Delta^-, \lambda_{\mathcal{N}}^-, Q_F, Q_S \rangle$. Then:

$$[[Q]]_{SDR} = [[\langle \mathcal{G}, \mathcal{R}^-, v_s \rangle]]_{SSR}.$$

Boolean matrix decomposition of $\mathcal{N}^-$ is obtained by transposing the matrices in the decomposition of $\mathcal{N}$.

4 A Semiring Framework for Single-Source RPQs

In this section, we present our main contribution: a generalized semiring-based framework for evaluating single-source 2-RPQs. The original LARPQ algorithm operates over the Boolean semiring to solve reachability. We show that by replacing the Boolean semiring with other algebraic structures, the same algorithmic framework can solve different path-aggregation problems.

4.1 Generalized Algorithm

The key insight is that the LARPQ algorithm can be abstracted to work over any semiring equipped with appropriate operations. Algorithm 2 presents the generalized version.

Input: Semiring $\mathcal{S} = \langle D_{out}, D_{in_1}, \mathbb{B}, \oplus, \otimes, \mathbf{0} \rangle$, optional index unary operator I, 2-NFA $\mathcal{N} = \langle Q, \Sigma, \Delta, \lambda_{\mathcal{N}}, Q_S, Q_F \rangle$, graph $\mathcal{G} = \langle V, E, L, \lambda_{\mathcal{G}} \rangle$, source $v_s \in V$
Output: Result depending on semiring instantiation

1. $\{N^a\} \leftarrow$ Boolean decomposition of the $\mathcal{N}$ adjacency matrix
2. $\{G^a\} \leftarrow$ Boolean decomposition of the $\mathcal{G}$ adjacency matrix
3. $M \leftarrow \mathbf{0}$, $P \leftarrow \mathbf{0}$
4. For each $q \in Q_S$: $M_{qv_s} \leftarrow \mathbf{1}$
5. While $M \neq \mathbf{0}$:
   - If I defined: $M \leftarrow I(M)$
   - $P \leftarrow P \oplus M$
   - $M \leftarrow \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} ((M^T \otimes N^a)^T \otimes G^a)$
6. Return Export results from P based on Q_F

Fig. 2: Generalized Single-Source 2-RPQ using Semirings

The generalized algorithm differs from the original in several key aspects:

- Matrix operations ($\oplus$, $\otimes$) are performed over an arbitrary semiring $\mathcal{S}$ having $D_{in_2} = \mathbb{B}$;

- An optional index unary operator I can be applied to filter or transform path information;
- The final result format depends on the semiring instantiation.

We now present two important instantiations of this framework.

4.2 Instantiation I: Boolean Semiring for Reachability

By instantiating $\mathcal{S}$ with the Boolean semiring $\langle \mathbb{B} = \{0,1\}, \mathbb{B}, \mathbb{B}, \vee, \wedge, 0 \rangle$ and completely omitting the index unary operator, we recover the original LARPQ algorithm for reachability (Algorithm 1). The matrix entries are Boolean values, where 1 indicates reachability. The mask operation $\langle \neg P \rangle$ ensures each vertex is processed only once.

The result is a Boolean vector P^F where $P^F_v = 1$ if and only if vertex v is reachable from the source via a path satisfying the regular language constraint.

4.3 Instantiation II: Path Semiring with Index Operators for All Paths

For enumerating all simple paths or all trails, we use the previously introduced path semiring $\langle RPaths, RPaths, \mathbb{B} = \{0,1\}, \cup, \mathbf{fst}, \varnothing \rangle$ combined with appropriate index unary operators.

When $I = I_{Simple}$, the algorithm enumerates all simple 2-paths satisfying the query. When $I = I_{Trail}$, it enumerates all trails. The key difference from reachability is that the frontier matrix M now stores actual path information (as sequences of vertices) rather than Boolean values. The index unary operator filters extensions that would violate the path semantics:

- I_{Simple} prevents extending a path with a vertex already present in the path (except the starting vertex), ensuring simplicity;
- I_{Trail} prevents extending a path with an edge already traversed, ensuring the result is a trail.

The algorithm maintains the BFS structure: on iteration n, M_{qv} contains all valid paths of length n from v_s to v that correspond to paths from some $q_s \in Q_S$ to q in the automaton.

Figure 3 illustrates an example of the algorithm step for the all-paths semantics.

[Figure: Placeholder]Algorithm step visualization for path semiring

Fig. 3: The algorithm step for graph and automaton for regular expression a^*b using path semiring.

4.4 Computational Complexity

We analyze the computational complexity of the generalized algorithm.
Complexity. Let $n = |V|$ be the number of vertices, $m = |E|$ be the number of edges, $k = |L|$ be the number of distinct labels, and $q = |Q|$ be the number of automaton states. The algorithm requires $O(q \cdot nz(G) \cdot \text{iterations})$ operations defined in semiring, where $nz(G)$ is the number of non-zero entries in the Boolean matrix decomposition.

4.5 Proof of Correctness

We prove the correctness of the generalized semiring-based algorithm.

Frontier Invariant. After the t-th iteration of the main loop, for each automaton state $q \in Q$ and vertex $v \in V$:

$$M_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } t \text{ satisfying the query}\}.$$

Accumulator Invariant. After the t-th iteration:

$$P_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } \leq t \text{ satisfying the query}\}.$$

We now provide a formal proof of correctness using invariant-based reasoning.

Invariant 1 (Frontier). After iteration t ($t \geq 0$), for each automaton state $q \in Q$ and vertex $v \in V$:

$$M_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length exactly } t \text{ such that } \omega(\pi) \in L(\mathcal{N})\}.$$

Invariant 2 (Accumulator). After iteration t ($t \geq 0$), for each $q \in Q$ and $v \in V$:

$$P_{qv}^{(t)} = \{\pi \mid \pi \text{ is a 2-path from } v_s \text{ to } v \text{ of length } \leq t \text{ such that } \omega(\pi) \in L(\mathcal{N})\}.$$

When an index unary operator I is applied, the invariant is modified to consider only paths satisfying the semantic constraints enforced by I.

Proof of Invariant 1 (Frontier). We prove by induction on t.

Base case ($t = 0$): Initially, $M^{(0)}$ contains the empty path from v_s to v_s for each starting state $q \in Q_S$, represented by the identity element $\mathbf{1}$ of the semiring. Formally:

$$M_{qv}^{(0)} = \begin{cases} \mathbf{1} & \text{if } q \in Q_S \text{ and } v = v_s \\ \mathbf{0} & \text{otherwise} \end{cases}$$

This matches the definition in Invariant 1 for paths of length 0.

Inductive step: Assume Invariant 1 holds after iteration t. Consider iteration $t + 1$. The update step computes:

$$M^{(t+1)} = \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} (N^a)^T \otimes M^{(t)} \otimes G^a.$$

For a fixed label a, the product $(N^a)^T \otimes M^{(t)}$ extends each path in $M^{(t)}$ by one automaton transition labeled a. Specifically, if $M^{(t)}_{q'v'}$ contains a path π, then $((N^a)^T \otimes M^{(t)})_{qv'}$ contains the extended path for each predecessor state q with a transition $q \xrightarrow{a} q'$ in $\mathcal{N}$.

Multiplying by G^a then extends these paths by one graph edge labeled a. The result contains all paths of length $t+1$ from v_s to each vertex v that correspond to paths in the automaton from some starting state to q.

Taking the union (via $\bigoplus$) over all labels a combines all possible extensions, yielding exactly the set of paths of length $t+1$ satisfying the query. This establishes Invariant 1 for iteration $t+1$.

When an index unary operator I is applied, it filters or transforms the paths in $M^{(t+1)}$ according to the semantic constraints. The correctness of filtering follows directly from the definitions of I_{Simple} and I_{Trail}.

Proof of Invariant 2 (Accumulator). Invariant 2 follows directly from Invariant 1 and the accumulation step:

$$P^{(t+1)} = P^{(t)} \oplus M^{(t+1)}.$$

By the induction hypothesis, $P^{(t)}$ contains all paths of length $\leq t$. Adding $M^{(t+1)}$ (which contains paths of exactly length $t+1$ by Invariant 1) yields all paths of length $\leq t+1$.

Theorem (Correctness). Let $\mathcal{Q} = \langle \mathcal{G}, \mathcal{R}, v_s \rangle$ be a 2-RPQ and let the algorithm terminate after T iterations. Then:

1. For Boolean semiring: $P^F_v = 1$ iff $v \in [[Q]]_{SSR}$;
2. For path semiring with I_{Simple}: the result contains exactly all 2-simple paths in $[[Q]]_{SSASP}$;
3. For path semiring with I_{Trail}: the result contains exactly all 2-trails in $[[Q]]_{SSAT}$.

Proof. By Invariant 2, after termination (when $M = \mathbf{0}$), P contains all paths from v_s to each vertex that satisfy the query. The export step multiplies by F, which selects only paths ending in final automaton states.

For Boolean semiring, $P_{q_F v} = 1$ for some $q_F \in Q_F$ iff there exists a path from v_s to v whose label is accepted by the automaton. This is precisely the definition of reachability.

For the path semiring, I_{Simple} ensures that only simple paths remain (no vertex repetition), and I_{Trail} ensures only trails remain (no edge repetition). The index operators are applied at each step, filtering out non-compliant extensions immediately, guaranteeing correctness.

5 Implementation Details

We have implemented the proposed algorithms within the LAGraph project [**?**], a collection of linear algebra-based graph processing algorithms built on top of

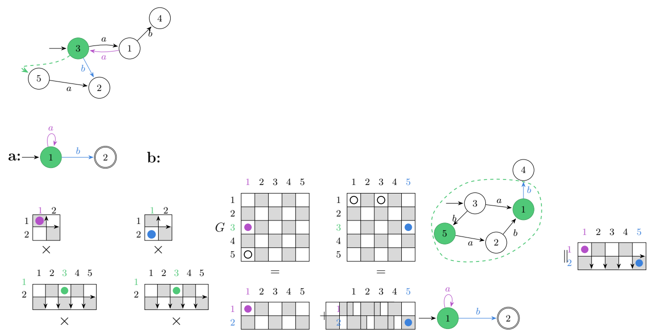


Fig. 4: Frontier update with path semantics: (a) simple paths filter, (b) trails filter.

SuiteSparse:GraphBLAS [**?**]. This section describes the key implementation aspects.

Secondly, the actual implementation involves an additional semiring due to the asymmetric nature of the specific definition. The original semiring $\textsc{Paths} = \langle \text{R}\textsc{Paths}, \text{R}\textsc{Paths}, \mathbb{B} = \{0,1\}, \cup, \mathbf{fst}, \varnothing \rangle$ utilizes $\mathbf{fst} : \text{R}\textsc{Paths} \times \mathbb{B} \to \text{R}\textsc{Paths}$.

However, it turns to be handy to introduce another semiring $\overline{\textsc{Paths}} = \langle \text{R}\textsc{Paths}, \mathbb{B} = \{0,1\}, \text{R}\textsc{Paths}, \cup, \mathbf{snd}, \varnothing \rangle$ utilising $\mathbf{snd} : \mathbb{B} \times \text{R}\textsc{Paths} \to \text{R}\textsc{Paths}$ defined as ??.

If we assume these two semirings and obtain two multiplicative operations $\otimes$ and $\overline{\otimes}$ we can observe it holds that.

$$(A \otimes B)^T = (B^T \overline{\otimes} A^T)$$

for matrices A and B over D_{out}.

That is to say, using this observation we can omit the expensive product transposition on line ?? in Algorithm 2 for the sake of the cheaper $(N^a)^T$ operation:

$$((M^T \otimes N^a)^T \otimes G^a) \rightsquigarrow ((N^a)^T \overline{\otimes} M \otimes G^a)$$

.

Apart from that, SuiteSparse:GraphBLAS impose some restrictions on user-defined semirings. Mostly important, user-defined semirings should have been of fixed size in order to stick with the builtin automatic allocators. To this end, implementation of Paths using SuiteSparse:GraphBLAS relies upon external memory management when paths become too long or too many paths appear in a single M_{qv} RPaths element. We introduce two constants: L_{fast} and C_{fast} for controlling maximum path length and maximum RPaths elements cardinality without external memory management.

If a path $\pi \in M_{qv}$ having $|\pi| > L_{fast}$ occurs a new linked list representing the lengthy path is allocated in a supplementary memory arena. Similarly, if $|M_{qv}| > C_{fast}$ for some q, v, an extensible linked list.

These two values essentially offer a trade-off between peak memory consumption and execution speed. On the practical side, it is better these constant need to be adjusted per each dataset individually.

Finally, we ship the semiring-based algorithm implementation as a standalone executable file. A preprocessed graph decomposed as a set of boolean Matrix-Market matrices, a set of regular path query patterns described using SPARQL-like query languages are consumed as input and executed one-by-one. The CLI flag ?? `-semantics` can be used to switch between all simple and all trails path semantics. At the same time, `-fast-path-length` controls L_{fast} and `-fast-path-length` defines the C_{fast} constant.

!!!Algorithm Implementation Details!!!

6 Evaluation

We evaluate the proposed solution on three semantics (simple paths, shortest paths, and trails) using real-world knowledge graphs. The experiments are designed to answer the following research questions.

- **RQ1:** How does our implementation compare with production graph database systems?
- **RQ2:** What is the performance behavior across different query types and graph structures?

Experiments are conducted on a workstation equipped with a Ryzen 9 7900X (4.7 GHz, 12 cores), 128 GB of DDR5 RAM, running Ubuntu 22.04. The timeout is set to 60 seconds per query. Each query is executed five times: one warm-up run followed by four measurement runs.

Competitors We compare the performance of the proposed semiring-based LARPQ algorithm against state-of-the-art graph database management systems.

- **MillenniumDB** or MDB (version for rpq challenge) [14]: a persistent, open-source graph database with state-of-the-art RPQ performance and natural support of all three path semantics.
- **FalkorDB** (v4.18.5, formerly RedisGraph [7]): an open-source in-memory property-graph database using GraphBLAS. Among the three path semantics, only all-trails is naturally expressed in FalkorDB, thus only respective queries will be measured.

For all competitors, we use their respective query languages to specify queries. The time required for query processing (including parsing and optimization) is included in the measurements. All data is preloaded into storage, so graph preprocessing time is excluded from the results.

Dataset We used three different graphs and their respective sets of queries.

Wikidata [12]: 610 million edges, 91 million vertices, and 1 400 distinct labels. A total of 578 queries were filtered from a dump of 660 queries.

Yago-2S: 46 million edges, 7 million vertices, and 42 distinct labels. Seven complex queries were taken from [2].

RPQBench [15]: a synthetic dataset with 150 million edges, 57 million vertices, and 9 distinct labels. It includes 20 different query types, 10 queries were generated for each.

6.1 Experimental Results

We measure the total time for all queries, the mean and median time over all queries, and the mean and median per-query speedup over the proposed solution. A summary for all three graphs is presented in Tables 1, 3, and 2. Per-query results for Wikidata represented in 5. Only queries that were successfully evaluated for all competitors are included in these statistics. In figure 5d we additionally show queries that successfully evaluated for MillenniumDB and our solution, but failed in different reasons for FalkorDB.

Table 1: Performance Summary on Wikidata (time in milliseconds)

Metric	Simple		Shortest		Trails		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**
Total time	**224 682.1**	1 111 429.4	**205 629.8**	780 011.1	152 128.6	921 506.4	**81 519.5**
Mean time	**428.8**	2 121.0	**392.4**	1 488.6	312.4	1 892.2	**167.4**
Median time	**1.7**	35.6	**2.0**	27.9	**1.4**	36.9	4.5
Mean Speedup	**1.0**	0.6	**1.0**	0.8	**1.0**	0.5	0.9
Median Speedup	**1.0**	0.1	**1.0**	0.1	**1.0**	0.1	0.7

Table 2: Performance Summary on RPQ-bench (time in milliseconds)

Metric	Simple		Shortest		Trails		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**
Total time	**857.7**	2835.7	**890.0**	3 253.3	**849.5**	3 256.5	1 685.3
Mean time	**42.9**	141.8	**44.5**	162.7	**53.1**	203.5	105.3
Median time	0.3	**0.2**	0.3	**0.1**	0.3	**0.2**	**0.2**
Mean Speedup	1.0	**1.5**	1.0	**2.1**	1.0	**1.5**	1.2
Median Speedup	1.0	**1.8**	1.0	**2.4**	1.0	**1.7**	1.5

The evaluation shows that execution time strongly depends on the particular query and the choice of start and target vertices. We define a query as *simple* for a system if it requires relatively little time to evaluate, and *hard* if, conversely, it requires a relatively large amount of time.

Table 3: Performance Summary on YAGO-2S (time in milliseconds)

Metric	Simple[1]		Shortest		Trails[1]		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**[2]
Total time	**104.8**	215.3	**1 194.6**	1 955.7	**66.4**	478.7	—
Mean time	—	—	**170.7**	279.4	—	—	—
Median time	—	—	**167.2**	268.3	—	—	—
Mean Speedup	—	—	**1.0**	0.5	—	—	—
Median Speedup	—	—	**1.0**	0.6	—	—	—

[1] Only two queries executed successfully for all competitors.
[2] Timeout for all queries.

For RPQ-Bench (Table 2), the total and mean execution times of our solution are more than three times better than those of the competitors across all semantics. However, the median time and per-query speedup favor the competitors. This indicates that the dataset contains a relatively large number of simple queries, while our solution exhibits a significant advantage on harder queries.

This observation is consistent with the results presented in Figure 5, where MillenniumDB outperforms our solution on simple queries but is slower on difficult ones. The tail of queries that are challenging for competing solutions is thinner for LARPQ than for the competitors, as illustrated in Figure 5c: although FalkorDB achieves the best total and median times, its mean time is worse than that of our solution.

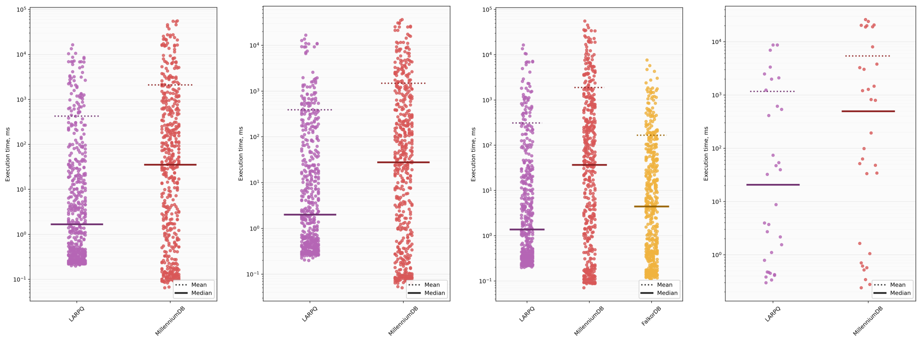
In the YAGO-2S dataset, the shortest-path semantics is the only one for which all seven queries completed successfully across all competing systems. For the simple-paths and trails semantics, many queries timed out. Nevertheless, for queries that completed successfully, our solution demonstrates better performance than the competitors.

A detailed analysis of which queries are *simple* and which are *hard* for our solution yields results similar to those reported in [5], indicating that the proposed generalization significantly inherits the properties of the basic version. Specifically, LARPQ outperforms competitors on queries with complex patterns over labels corresponding to many edges (e.g., $l_1/l_2/(l_3 \mid l_4)^*$), while on queries with simple patterns such as l_1^*/l_2, it shows no performance advantage, and MillenniumDB may be a better choice.

Finally, we conclude that LARPQ demonstrates reasonable performance in all cases and is a better choice for complex queries.

7 Related Work

Regular path queries have been extensively studied in the context of graph databases. In this section, we discuss related work on linear algebra-based graph analytics, RPQ evaluation techniques, and semiring-based path problems.



(a) Simple paths (b) Shortest paths (c) Trails (common for all competitors) (d) Trails (FalkorBD failed)

Fig. 5: Performance on Wikidata

Linear Algebra-Based Graph Analytics. The GraphBLAS initiative [**?**,**?**] established a standardized interface for expressing graph algorithms using linear algebra operations. LAGraph [**?**] provides a collection of graph algorithms built on GraphBLAS, including BFS, PageRank, and triangle counting. Our work extends this ecosystem with RPQ algorithms.

The use of sparse matrices for graph representation has proven effective for large-scale graphs. Real-world graphs exhibit sparsity, allowing for memory-efficient storage and fast computation using formats like CSR/CSC [**?**]. The GraphBLAS specification supports arbitrary semirings, enabling the expression of algorithms that go beyond traditional numeric computations.

Recent work has explored linear algebra approaches for various graph problems. However, RPQ evaluation in this paradigm remains underexplored, with only a few prior works [**?**,**?**] addressing this problem.

RPQ Evaluation Methods. Traditional approaches to RPQ evaluation include:

- **Automaton-based methods:** Construct the product automaton of the graph and the query NFA, then perform traversal [**?**]. These methods can be combined with indexing techniques for improved performance.
- **BFS-based methods:** Perform breadth-first search guided by the query automaton [**?**]. The original LARPQ algorithm [**?**] falls into this category.
- **Matrix-based methods:** Translate the regular expression into matrix operations. The RPQ-matrix algorithm [**?**] uses this approach, representing adjacency matrices in compressed formats like k2-trees or CSR/CSC.

Our approach combines elements from both BFS and matrix-based methods, using the simultaneous traversal of graph and automaton structures while leveraging sparse matrix operations for efficiency.

Different query semantics have been studied in the literature. The all-shortest-paths semantics is addressed in [?], while all-simple-paths and all-trails semantics are more challenging due to the potential for exponential result sizes. Our framework provides a unified approach for handling these different semantics.

Semiring-Based Path Problems. The algebraic approach to path problems has a long history. Kleene's algorithm [?] computes reachability using matrix operations over the Boolean semiring. This approach has been generalized to arbitrary semirings for solving various algebraic path problems [?,?].

The connection between semirings and path enumeration is well-established. The tropical semiring (min-plus) corresponds to shortest path problems, while the Boolean semiring corresponds to reachability. Our work extends this connection to include the path semiring for explicit path enumeration.

Provenance semirings [?] provide a framework for tracking the origin of query results. Our path semiring can be seen as a form of provenance tracking, where each result path is explicitly stored.

Context-free path querying (CFPQ) has also been approached using linear algebra [?]. While CFPQ is more expressive than RPQ, our work focuses on regular languages and different evaluation semantics.

Graph Database Systems. Modern graph database systems vary significantly in their RPQ support:

- **MillenniumDB** [?]: A persistent graph database with state-of-the-art RPQ performance, supporting various path semantics.
- **FalkorDB** [?] (formerly RedisGraph): An in-memory property-graph database that uses GraphBLAS for query evaluation. Supports a subset of regular path queries through Cypher.
- **Neo4j:** The most widely used graph database, supporting Cypher queries including limited forms of RPQ.
- **Blazegraph:** An RDF database with SPARQL support including property paths.

Our implementation differs from these systems in its focus on the linear algebra paradigm and its flexibility in supporting different semirings for various path semantics.

Table 4 summarizes the comparison of our approach with related systems.

[Table: Placeholder]Comparison of RPQ approaches

Table 4: Comparison of RPQ evaluation approaches. LA = Linear Algebra, AB = Automaton-based, BFS = Breadth-first search.

Our contributions differ from prior work in several key aspects:

- We provide a **unified framework** that can solve multiple path semantics using different semiring instantiations;
- We extend the original LARPQ algorithm beyond reachability to path enumeration;
- We leverage the **GraphBLAS standard** for high-performance sparse matrix operations.

8 Conclusion and Future Work

We have presented a semiring-based generalization of the linear algebra approach for evaluating two-way regular path queries. Source code and additional materials are available at https://github.com/SparseLinearAlgebra/la-rpq. Evaluation framework !!!. The experimental evaluation demonstrates that the proposed approach achieves competitive performance with state-of-the-art graph database systems while providing a more flexible framework for different path semantics.

Future work includes the use of additional semirings, such as the tropical semiring for shortest-path queries and counting semirings for path counting, as well as integration with property graphs and more expressive query languages (e.g., CRPQs). Another direction is a distributed implementation (e.g., using CombBLAS [?]) and acceleration by offloading matrix operations to GPUs (e.g., using GraphBLAST, cuBool, Spla [?]).

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